# 8 Radiative transfer: Solutions to Schwarzschild's equation

Reading for Day 8: Pierrehumbert Principles of Planetary Climate, Section 4.1-4.2.

## 8.1 Schwarzschild's equation recap

Recall from our previous day's notes that we can write Schwarzschild's equation for radiative transfer ignoring scattering as

$$\frac{dI_\nu}{dl} = \rho k_\nu (B_\nu - I_\nu) \text{ Schwarzschild's equation} \tag{8.1}$$

in length coordinates, or equivalently in optical path coordinates as

$$\frac{dI_\nu}{d\tilde{\tau}} = B_\nu - I_\nu, \tag{8.2}$$

or optical depth coordinates as

$$\mu \frac{dI_\nu}{d\tau} = I_\nu - B_\nu, \tag{8.3}$$

where $\mu = \cos\theta$, with $\theta$ the zenith angle between the light path and the vertical in the atmosphere. Moving forward, we'll use optical path coordinates and drop the tilde for simplicity (and I'll be clear when we're using optical depth instead).

## 8.2 Solutions to Schwarzschild's equation

### 8.2.1 Pure absorption: Beer's law

In the case where the medium is purely absorbing, $B_\nu = 0$.

*Quick activity* (complete in groups of 2-3): Integrate Schwarzschild's equation to solve for the radiance as a function of optical path. You should find

$$I_\nu = I_\nu(0) e^{-\tau} \quad \text{Beer's law.} \tag{8.4}$$

Thus, in the case of pure absorption the optical path $\tau$ can be interpreted as the e-folding scale of the radiance. If $\tau \gg 1$, then the region is optically thick and $I_\nu \approx 0$. If $\tau \ll 1$, then the region is optically thin and $I_\nu \approx I_\nu(0)$.

Beer's law has a wide range of applications, especially regarding the propagation of sunlight through Earth's atmosphere, as the thermal radiance of Earth is quite small in the optical (and thus we can well approximate $B_\nu \approx 0$ in Schwarzschild's equation). We discuss one such application to the slant geometry in Section 8.3.

### 8.2.2 Isothermal atmosphere

We previously considered atmospheres with no emission to derive Beer's law for transmission, but now we will consider atmospheres with emission. First, let us consider an isothermal medium where $B_\nu$ is constant. We can change variables to simplify our derivation, using

$$\begin{aligned} \chi &= I_\nu - B_\nu \\ d\chi &= dI_\nu. \end{aligned} \tag{8.5}$$

Thus, we can write Schwarzschild's equation with the change of variable as

$$\frac{d\chi}{d\tau} = -\chi. \tag{8.6}$$

Integrating, we find

$$\chi = \chi_0 e^{-\tau}. \tag{8.7}$$

Plugging back in for our change of variables, we find the isothermal solution of Schwarzschild's equation:

$$I_\nu = B_\nu + (I_\nu(0) - B_\nu)\, e^{-\tau}. \tag{8.8}$$

Let's consider some limits of this isothermal solution. When the optical path is very small ($\tau \ll 1$), the radiance is the same before and after entering the slab ($I_\nu \approx I_\nu(0)$), as in the case of Beer's law above. When the optical path is very large ($\tau \gg 1$), $I_\nu \approx B_\nu$ – that is, an infinitely opaque isothermal medium acts as a blackbody. Finally, if $I_\nu(0) = 0$, then $I_\nu = B_\nu(1 - e^{-\tau})$, and the radiance $e$-folds toward a blackbody with increasing optical thickness.

### 8.2.3 General solution

There of course is also a general solution to Schwarzschild's equation ignoring scattering. To derive it, we can use an integrating factor $e^\tau$ such that

$$\frac{d}{d\tau}(e^\tau I_\nu) = e^\tau B_\nu, \tag{8.9}$$

and thus

$$e^{\tau'} I_\nu|_0^\tau = \int_0^\tau e^{\tau'} B_\nu d\tau'. \tag{8.10}$$

One can then rearrange to find the general solution to Schwarzschild's equation,

$$I_\nu(\tau) = I_\nu(0)e^{-\tau} + \int_{\tau=0}^{\tau} e^{\tau'-\tau} B_\nu d\tau. \tag{8.11}$$

Note that the first half of the right hand side is simply Beer's law, and the second half is the additional contribution from thermal emission.

*Quick activity* (complete in groups of 2-3): Assume $B_\nu$ = constant. Which solution do you recover from the general solution, and why?

## 8.3 Application of Beer's law: slant vs. normal optical path

Beer's law can be applied to transmission geometry (i.e., the propagation of slant rays through an atmosphere – this is relevant both for exoplanet observations as well as stellar occultations of Earth and Solar System objects), as in most cases we can ignore the atmospheric emission contribution to the transit depth. This then allows us to relate the observed flux deficit to an optical path, and thus a region in the atmosphere that is probed via transmission spectroscopy, given that regions with $\tau > 1$ are opaque to transmitted starlight.

Let us consider a *slant* optical path through the atmosphere $\tau$ – this slant optical path is significantly larger than the typical optical depth. Figure 8.1 shows the geometry of the slant

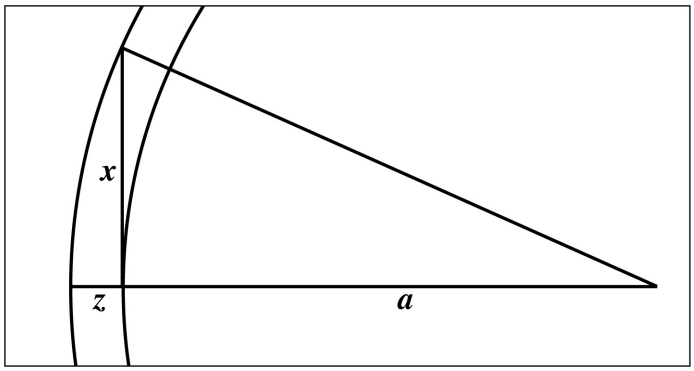



**Figure 8.1:** Geometry used to derive the slant optical path of transmission through an atmosphere. Figure adapted from Fortney (2005).

optical path (in the direction of $x$) relative to the vertical optical depth (in the direction of $z$). We will now follow the derivation in Fortney (2005)[30]. Recall that an isothermal atmosphere in hydrostatic equilibrium has a dependence of pressure $p$ and equivalently number density $n$ on height above a given level $z_0$ as

$$\begin{aligned} p(z) &= p(z_0)\ \exp\left(-\frac{(z-z_0)}{H}\right), \\ n(z) &= n(z_0)\ \exp\left(-\frac{(z-z_0)}{H}\right), \end{aligned} \tag{8.12}$$

where $H = RT/g$ is the pressure scale height, which is equal to the density scale height for an isothermal atmosphere. From Figure 8.1, we can note that

$$a^2 + x^2 = (a+z)^2 = a^2 + 2az + z^2. \tag{8.13}$$

If $2az \gg z^2$ (i.e., if the atmosphere is thin relative to the planetary radius), then $z \approx x^2/2a$. As a result, we can write the dependence of number density on $x$ as

$$n(x) = n_0\ \exp\left(-\frac{x^2}{2aH}\right). \tag{8.14}$$

If we integrate this dependence over $x$ to find the integrated density $N_H$ from horizon to horizon, we find

$$N_H = \int_{-\infty}^{\infty} n(x)dx = n_0 \frac{1}{2}\sqrt{\pi}\sqrt{2aH}\ \mathrm{erf}\left(\frac{x}{\sqrt{2aH}}\right)\Big|_{-\infty}^{\infty} = n_0\sqrt{2\pi aH}. \tag{8.15}$$

The ratio of the horizontally integrated density to the vertically integrated density is equal to the ratio of the horizontal to vertical optical path. This is

$$\frac{N_H}{N_V} = \frac{\tau_H}{\tau_V} = \sqrt{\frac{2\pi a}{H}}. \tag{8.16}$$

[30]This is derived specifically in the context of to exoplanets, but can be applied to any slant optical path, also relevant for Earth observation.

Because $a >> H$, this value is always much larger than one – for Earth it is $\sim 75$, and for Jupiter it is $\sim 128$. Generally speaking, due to the high slant optical path, transit spectra probe low pressures of $\approx 1$ mbar, with higher resolution observations probing even lower pressures in absorption lines. This also causes condensate clouds and hazes to have a large impact on the depth of transmission spectral features, as they move the continuum (deepest region that can be probed) to lower pressures (Fortney, 2005).

## 8.4 Practical application: transmission spectroscopy

### 8.4.1 Transmission spectroscopy

Transmission spectroscopy probes the atmospheres of exoplanets by studying the transmission (or filtering) of light from the host star through the limb of an exoplanet that appears to occult the host star. Note that the same physics also applies to occultations in our Solar System, either Solar occultations (where one probes the filtered sunlight through the object's limb) or stellar occultations (where one probes filtered starlight from a more distant star).

Figure 8.2 shows the geometry of a transit event, along with the geometry of the secondary eclipse that is used to derive planetary emission spectra (that we discussed in the last class). Transmission spectra are generally measuring small deviations (at most on the orders of

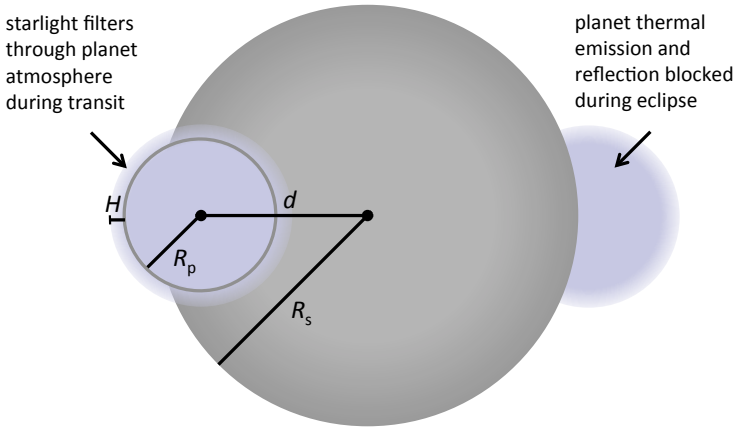



**Figure 8.2:** Geometry of transmission and emission observations. Here the stellar radius is $R_s$, planet radius is $R_p$, the separation of centers of planet and star is $d$, and the pressure scale height of the planet's atmosphere is $H$. Figure adapted from Kreidberg (2017).

parts per thousand) on top of the larger transit signal. Similarly to the transit method itself, transmission spectral signatures are larger for planets orbiting smaller stars. However, there is also a strong dependence of the transmission spectra signal on the atmospheric composition, temperature, and gravity of the orbiting planet, which we derive below.

The transmission spectral features are a small atmospheric contribution imprinted on top of the total transit depth. Note that the transit depth for the solid-body planet is $\delta = (R_p/R_s)^2$ (this can be derived by considering the ratios of the projected solid angles of the planet and host star). The transit depth including the atmospheric contribution is then

$$\delta_\lambda = \frac{(R_p + A_{H,\lambda})^2}{R_\star^2}, \tag{8.17}$$

where $A_{H,\lambda}$ is the apparent atmospheric height which is being probed. We can write the apparent atmospheric height as a number of scale heights, i.e.,

$$A_{H,\lambda} = nH = n\frac{RT}{g} \tag{8.18}$$

where $H$ is the (isothermal) pressure scale height. As a result, we can isolate the transmission spectral feature amplitude (i.e., the contribution due to the atmosphere alone) as

$$\delta_{\lambda,atm} = \frac{(R_p + nH)^2}{R_\star^2} - \frac{R_p^2}{R_\star^2} \approx \frac{2R_p nH}{R_\star^2} \approx \frac{2R_p nRT}{gR_\star^2}. \tag{8.19}$$

Typically, $n \approx 2$ for low spectral resolution observations of cloud-free atmospheres (Kreidberg, 2017). As a result, we expect the transmission spectral feature amplitudes to be larger for hotter and lower-gravity planets, which have a larger scale height. We also expect transmission spectral features to be larger for low mean molecular weight atmospheres, as decreasing mean molecular weight increases $R$, which also increases the scale height. Note that our derivation does not include the effects of clouds, which also mute spectral features by increasing the optical depth of the atmosphere at low pressures.

### 8.4.2 The case of WASP-43b

Figure 8.3 shows a transmission spectrum observed with the Hubble Space Telescope/Wide Field Camera 3 (HST/WFC3) (bottom) of the hot Jupiter WASP-43b. This

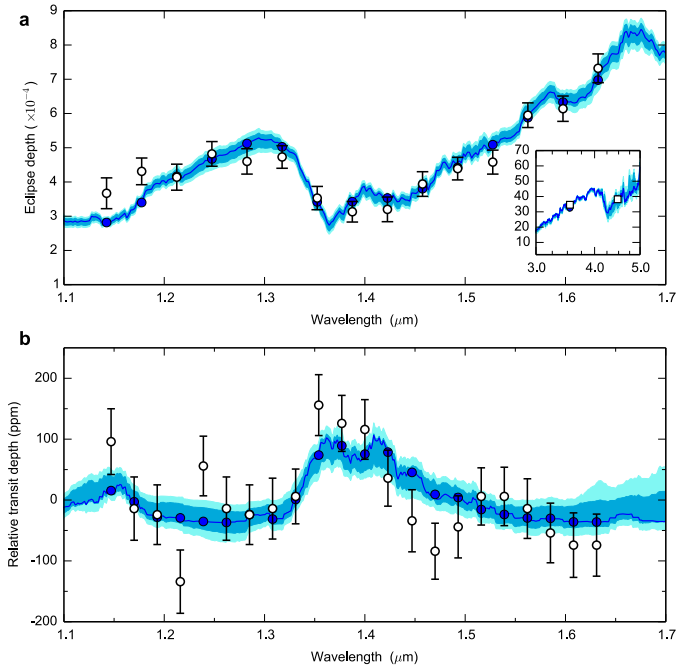



**Figure 8.3:** Emission spectrum (top) and transmission spectrum (bottom) of WASP-43b observed with the Hubble Space Telescope Wide Field Camera 3 instrument (inset on top shows observations with the Spitzer Space Telescope). Note the spectral feature (bump) in the transmission spectrum at 1.4 $\mu$m due to water vapour absorption, which also causes an absorption feature (dip) in the emission spectrum. Figure adapted from Kreidberg (2017).

shows the change in transit depth as a function of wavelength, i.e., isolating only the atmospheric contribution $A_H$ to the total transit depth. There is a bump in the spectrum at a wavelength of 1.4 $\mu$m. This bump corresponds to an increase in the effective transit depth, i.e., an increase in the effective size of the planet at 1.4 $\mu$m. As a result, this is an

absorption feature, as the planet absorbs more light at 1.4 $\mu$m than at other wavelengths in the instrumental range. Note that this bump in the transmission spectrum at 1.4 $\mu$m aligns well with a dip in the emission spectrum (top) at the same wavelength. Note that this dip in emission spectra similarly corresponds to an absorption feature due to the same species causing the bump in the transmission spectrum.

Given a transmission spectral feature at a given wavelength, astronomers can then infer the species that is causing this feature by comparing with model atmospheres. The wavelength-dependence of the atmospheric absorption is determined by the absorption cross section of a given species $\sigma$, which is related to opacity (equivalently absorption coefficient) $\kappa$ as

$$n\sigma_\lambda = \kappa_\lambda \rho, \tag{8.20}$$

where subscripts indicate a wavelength-dependence. Figure 8.4 shows the dominant cross-sections in a typical hot Jupiter atmosphere in chemical equilibrium at Solar metallicity, a temperature of 1500 K, and a pressure of 300 mbar.

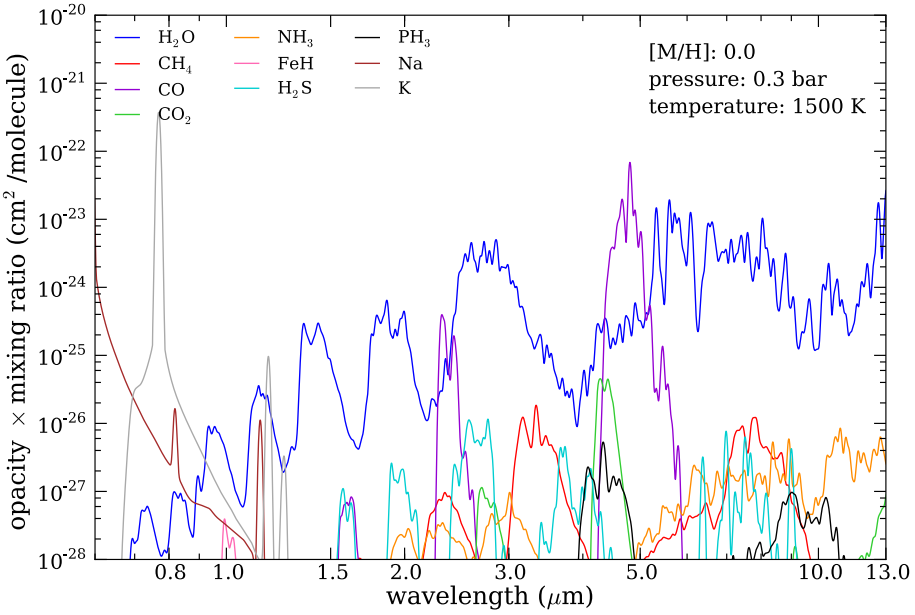



**Figure 8.4:** Cross sections of various species in a Solar composition atmosphere of a hot Jupiter at a pressure of 300 mbar. Note the strong features due to water vapor and carbon monoxide in the near-infrared. Figure adapted from Kreidberg (2017).